

Randy Tang

rtang0203@gmail.com | 732-666-2746 | U.S. Citizen

SUMMARY

Software engineer with 3+ years building **high-throughput, data-intensive production systems** at a global bank. Deep Python proficiency with hands-on experience designing and optimizing **real-time data pipelines** processing tens of millions of transactions daily. Strong **SQL** and data analysis skills, Cornell CS foundation in **statistics and algorithms**, and genuine interest in **equity markets and quantitative analytics**.

TECHNICAL SKILLS

Languages: Python (advanced), SQL (advanced), TypeScript | **Data & Infra:** Pandas, Postgres, SQLite, AMPS message queues (pub/sub), high-throughput pipelines, concurrency, Docker

Analytics & Markets: Large-scale data analysis (50-100GB datasets), back-testing, equity & crypto market data, performance optimization | **Tools:** Git, Flask, FastAPI, Grafana, LLM APIs

EXPERIENCE

Software Engineer — Bank of America

Aug 2022 – Present

Cirrus OTC Trade Reporting | Python, Pandas, Quartz, AMPS message queues

- Designed and optimized **high-throughput data pipelines** processing tens of millions of OTC transactions daily across the firm's trade reporting platform.
- Built a **reference-data framework**: pipeline of services via AMPS queues querying multiple external sources per trade event; handled retries, API calls, load balancing — extensible for new data sources.
- Developed T+1 **UTI validation service** ensuring trade-ID uniqueness across counterparties with automated corrections; processed millions of transactions/day.
- Parallelized and **automated a formerly manual** backreporting process, delivering large speed and throughput gains.
- Built transformer service (internal model → outgoing XML) and **DTCC response parser** to update transaction status in real time.
- Wrote Python/Pandas scripts to scrape, clean, and **analyze 50-100GB datasets** for trade reconciliation and reporting insights.

Data Analyst (Contract) — Microsoft

Mar 2022 – Aug 2022

Viva Topics | Python, Pandas, Jupyter

- Processed and analyzed large datasets in Python/Pandas used to train ML models; wrote data-cleaning and reporting scripts.

Engineering Intern — Northrop Grumman

Fall 2019, Summer 2020

NX, ANSYS

- Released 50+ engineering drawings incl. new designs and large assemblies; fully designed/drafted/released multiple flight and test components.
- Structural analyses for large assemblies and detailed fastener-stack analyses.

PROJECTS

- Market Data Dashboard** (Grafana, Twilio, Postgres, Docker) — Pulls **equity and crypto market data** (open interest, funding rates, price) from exchange APIs; stores in **Postgres**, visualizes in **Grafana** dashboards with configurable SMS alerts on signal triggers via Twilio.
- Polymarket Scanner** (Flask, Discord API, SQLite) — Scans prediction markets for **large trades and unusual activity patterns**; applies signal detection logic and alerts to Discord for real-time monitoring.
- Benchmark Agency Platform** (Next.js, React, TypeScript, Tailwind, Recharts, Supabase, Python, SQLite, Google Places API, Anthropic API, BeautifulSoup) — Multi-tenant **financial-intelligence dashboard** (revenue trends, equipment P&L, staff comp, AI-generated recommendations) with row-level security. Built lead-gen **data pipeline** (8 idempotent stages, 20 tests) discovering businesses via Google Places API and scoring via LLM.

EDUCATION

Cornell University, College of Engineering — Ithaca, NY

Sep 2017 – Dec 2021

B.S. Computer Science & B.S. Mechanical Engineering | GPA 3.7

- CUAIR (Cornell Unmanned Aircraft) — autonomous systems with obstacle avoidance, object classification; top team at AUVSI SUAS.